Activity No:	
Activity Title: Problem: Temperature Converter	
Course Code: CPE007	**Program:** Computer Engineering
Course Title: Programming Logic and Design	**Date Performed: October 22, 2026**
Section: CPE11S2	**Date Submitted: October 22, 2026**
Name(s): Zachary Truman A. Cortez	**Instructor: Sir Lusung**

Objectives

- **This activity aims to develop programming skills using functions.**

Intended Learning Outcomes (ILOs):

2.1 Understand how to construct programs modularly from small pieces called functions.

2.2 Implement common math functions available in the C standard library

2.3 Create new functions

Instructions

Activities:

Note: *Perform each of these steps for the below problems:*

- *Read the problem statement*
- *Write a C program*
- *Test, Debug and execute the C program*
- *Copy and paste the code and screenshot of the output then submit it using doc, docx, and pdf formats only.*

1. Create a program that defines a function to compute for the area of a cube. The formula of the area of the cube is given as A = s * s * s.
2. Define a function hypotenuse that calculates the length of the hypotenuse of a right triangle when the other two sides are given. Use this function in a program to determine the length of the hypotenuse for each of the following triangles. The function takes two arguments of type double and return the hypotenuse as a double.
3. Implement the following integer functions:
    1. Function celsius returns the Celsius equivalent of a Fahrenheit temperature.
    2. Function fahrenheit returns the Fahrenheit equivalent of a Celsius temperature.
    3. Use these functions to write a program that prints charts showing the Fahrenheit equivalents of a Celsius temperatures from 0 to 100 degrees, and the Celsius equivalents of all Fahrenheit temperatures from 32 to 212 degrees. Print the outputs in a neat tabular format that minimizes the number of lines of output while remaining readable.

Output/Answer

1.
CODE:

```
#include <iostream>
#include <iomanip>
#include <cmath>
using namespace std;

// Function prototype
double cubeArea(double s);

int main()
{
```

```
    double s;
          // side length entered by the user

    cout << "Enter the side of the cube: ";
    cin >> s;

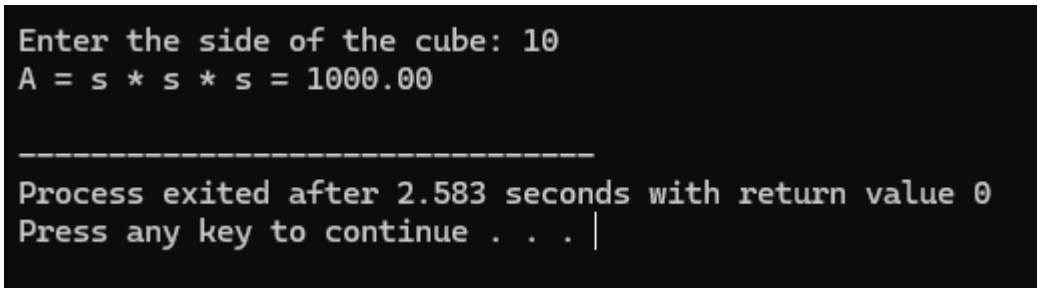
    // Print results with 2 digits after the decimal point
    cout << fixed << setprecision(2);

    // Call cubeArea() and display the returned value
    cout << "A = s * s * s = " << cubeArea(s) << endl;

    return 0;   // program ended successfully
}

// Function definition
// Parameter s: side length of the cube
// Returns: s raised to the power 3
double cubeArea(double s)
{
    return pow(s, 3);
}
```

OUTPUT:



2.

CODE:

```
#include <iostream>
#include <iomanip>
#include <cmath>
using namespace std;

// Function prototype
double hypotenuse(double side1, double side2);

int main()
{
    double side1, side2;   // the two sides entered by the user
    char again;

    // Show two digits after the decimal point
    cout << fixed << setprecision(2);

    do
```

```
    {
        // Ask the user for the two sides of the right triangle
        cout << "Enter side 1: ";
        cin >> side1;
        cout << "Enter side 2: ";
        cin >> side2;

        // Call hypotenuse() and display the returned value
        cout << "The hypotenuse is " << hypotenuse(side1, side2) << endl;

        // Ask whether to test another triangle
        cout << "Another triangle? (y/n): ";
        cin >> again;
        cout << endl;

    } while (again == 'y' || again == 'Y');

    return 0;
}

// Function definition
// Parameters: side1, side2 - lengths of the two sides that form the right angle
// Returns: the hypotenuse, by the Pythagorean theorem c = sqrt(a^2 + b^2)
double hypotenuse(double side1, double side2)
{
    return sqrt(pow(side1, 2) + pow(side2, 2));
}
```

OUTPUT:

```
Enter side 1: 10
Enter side 2: 10
The hypotenuse is 14.14
Another triangle? (y/n): y

Enter side 1: 34
Enter side 2: 53
The hypotenuse is 62.97
Another triangle? (y/n): y

Enter side 1: 200
Enter side 2: 2
The hypotenuse is 200.01
Another triangle? (y/n): n


--------------------------------
Process exited after 42.04 seconds with return value 0
Press any key to continue . . .
```

3.

CODE:

```
#include <iostream>   // cout, endl
#include <iomanip>    // setw
#include <cmath>      // floor
using namespace std;

// Function prototypes
int celsius(int f);      // Fahrenheit in, Celsius out
int fahrenheit(int c);   // Celsius in, Fahrenheit out

int main()
{
   // Table headers: each table has two columns
   cout << setw(5) << "C" << setw(6) << "F" << " |"
        << setw(5) << "C" << setw(6) << "F" << " |"
        << setw(5) << "F" << setw(6) << "C" << " |"
        << setw(5) << "F" << setw(6) << "C" << endl;
   cout << "------------+------------+------------+-----------" << endl;

   // The longest table (Fahrenheit 32..122) has 91 rows, so 91 lines
   // are enough for all four tables
   for (int i = 0; i <= 90; i++)
   {
      // Table 1: Celsius 0..50 (rows 0 to 50)
      if (i <= 50)
         cout << setw(5) << i << setw(6) << fahrenheit(i);
      else
         cout << setw(5) << "" << setw(6) << "";   // blank to keep alignment
      cout << " |";

      // Table 2: Celsius 51..100 (rows 0 to 49)
      if (i <= 49)
         cout << setw(5) << 51 + i << setw(6) << fahrenheit(51 + i);
      else
         cout << setw(5) << "" << setw(6) << "";
      cout << " |";

      // Table 3: Fahrenheit 32..122 (rows 0 to 90)
      cout << setw(5) << 32 + i << setw(6) << celsius(32 + i) << " |";

      // Table 4: Fahrenheit 123..212 (rows 0 to 89)
      if (i <= 89)
         cout << setw(5) << 123 + i << setw(6) << celsius(123 + i);

      cout << endl;
   }

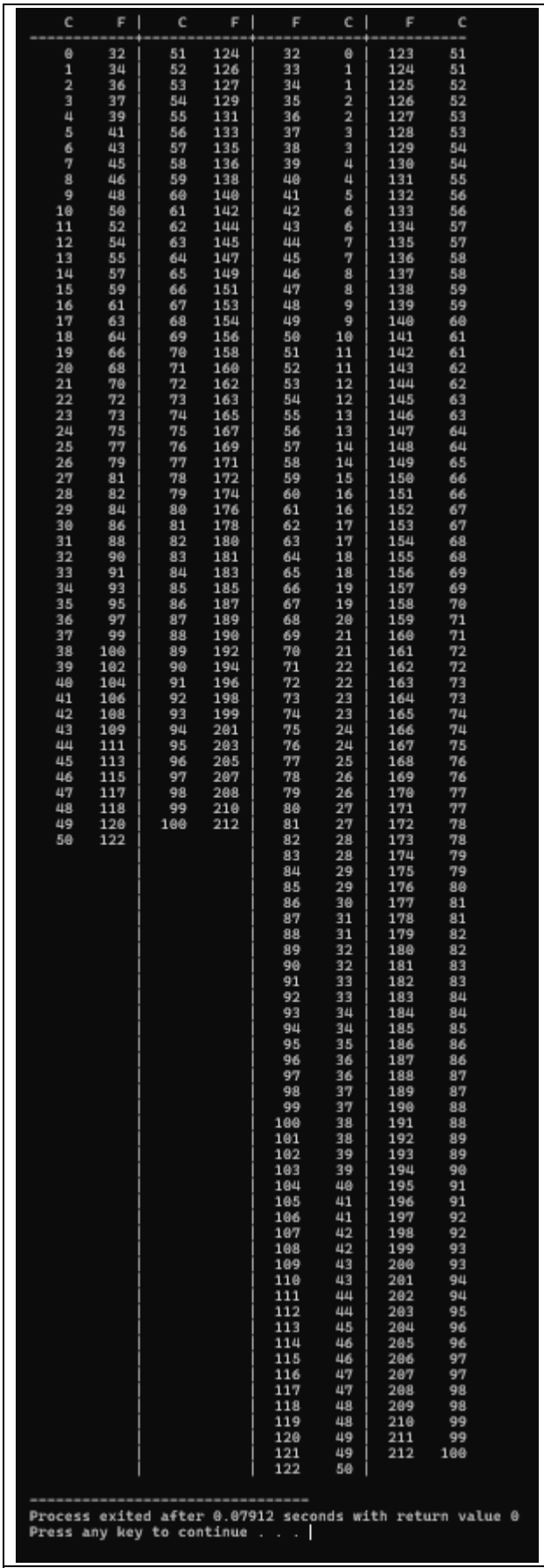
   return 0;
}

// Returns the Celsius equivalent of a Fahrenheit temperature
// Formula: C = 5/9 * (F - 32), rounded to the nearest integer
```

```
int celsius(int f)
{
    return (int) floor(5.0 * (f - 32) / 9.0 + 0.5);
}

// Returns the Fahrenheit equivalent of a Celsius temperature
// Formula: F = 9/5 * C + 32, rounded to the nearest integer
int fahrenheit(int c)
{
    return (int) floor(9.0 * c / 5.0 + 32 + 0.5);
}
```

OUTPUT:



C	F	C	F	F	C	F	C
0	32	51	124	32	0	123	51
1	34	52	126	33	1	124	51
2	36	53	127	34	1	125	52
3	37	54	129	35	2	126	52
4	39	55	131	36	2	127	53
5	41	56	133	37	3	128	53
6	43	57	135	38	3	129	54
7	45	58	136	39	4	130	54
8	46	59	138	40	4	131	55
9	48	60	140	41	5	132	56
10	50	61	142	42	6	133	56
11	52	62	144	43	6	134	57
12	54	63	145	44	7	135	57
13	55	64	147	45	7	136	58
14	57	65	149	46	8	137	58
15	59	66	151	47	8	138	59
16	61	67	153	48	9	139	59
17	63	68	154	49	9	140	60
18	64	69	156	50	10	141	61
19	66	70	158	51	11	142	61
20	68	71	160	52	11	143	62
21	70	72	162	53	12	144	62
22	72	73	163	54	12	145	63
23	73	74	165	55	13	146	63
24	75	75	167	56	13	147	64
25	77	76	169	57	14	148	64
26	79	77	171	58	14	149	65
27	81	78	172	59	15	150	66
28	82	79	174	60	16	151	66
29	84	80	176	61	16	152	67
30	86	81	178	62	17	153	67
31	88	82	180	63	17	154	68
32	90	83	181	64	18	155	68
33	91	84	183	65	18	156	69
34	93	85	185	66	19	157	69
35	95	86	187	67	19	158	70
36	97	87	189	68	20	159	71
37	99	88	190	69	21	160	71
38	100	89	192	70	21	161	72
39	102	90	194	71	22	162	72
40	104	91	196	72	22	163	73
41	106	92	198	73	23	164	73
42	108	93	199	74	23	165	74
43	109	94	201	75	24	166	74
44	111	95	203	76	24	167	75
45	113	96	205	77	25	168	76
46	115	97	207	78	26	169	76
47	117	98	208	79	26	170	77
48	118	99	210	80	27	171	77
49	120	100	212	81	27	172	78
50	122			82	28	173	78
				83	28	174	79
				84	29	175	79
				85	29	176	80
				86	30	177	81
				87	31	178	81
				88	31	179	82
				89	32	180	82
				90	32	181	83
				91	33	182	83
				92	33	183	84
				93	34	184	84
				94	34	185	85
				95	35	186	86
				96	36	187	86
				97	36	188	87
				98	37	189	87
				99	37	190	88
				100	38	191	88
				101	38	192	89
				102	39	193	89
				103	39	194	90
				104	40	195	91
				105	41	196	91
				106	41	197	92
				107	42	198	92
				108	42	199	93
				109	43	200	93
				110	43	201	94
				111	44	202	94
				112	44	203	95
				113	45	204	96
				114	46	205	96
				115	46	206	97
				116	47	207	97
				117	47	208	98
				118	48	209	98
				119	48	210	99
				120	49	211	99
				121	49	212	100
				122	50		

```
--------------------------------
Process exited after 0.07912 seconds with return value 0
Press any key to continue . . .
```

Conclusion

In this activity, I learned how to divide a program into smaller modules called functions, using prototypes, definitions, and return values to keep the code organized and reusable. I also practiced using the math library functions in <cmath>, such

as pow, sqrt, and floor, to handle calculations like the cube's area, the hypotenuse, and temperature conversion. Writing and testing each program helped me understand how arguments are passed to functions and how a function's result is used in main, which made my programs easier to read, debug, and modify.